

3.1 Computer Network

A computer network is a group of computers linked to each other that enables them to communicate with each other and share their resources, data, and applications.

3.1.1 Types of Computer Networks

A computer network can be categorized by their size. A **computer network** is mainly of **four types** (as shown in Figure 3.1):

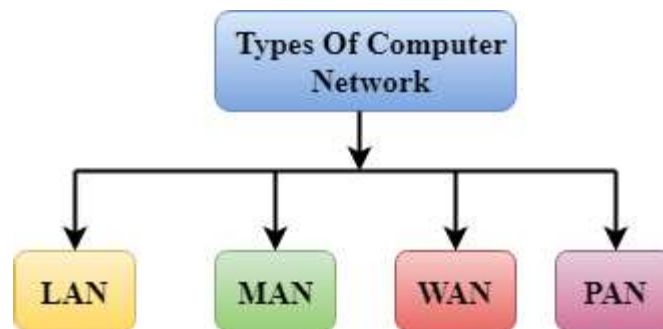


Fig. 3.1 Types of Computer Networks

- LAN (Local Area Network)
- PAN (Personal Area Network)
- MAN (Metropolitan Area Network)
- WAN (Wide Area Network)

LAN (Local Area Network)

Local Area Network is a group of computers connected to each other (as shown in Figure 3.2) in a small area such as building, office.

- LAN is used for connecting two or more personal computers through a communication medium such as twisted pair, coaxial cable, etc.
- It is less costly as it is built with inexpensive hardware such as hubs, network adapters, and ethernet cables.
- The data is transferred at an extremely faster rate in Local Area Network.
- Local Area Network provides higher security.

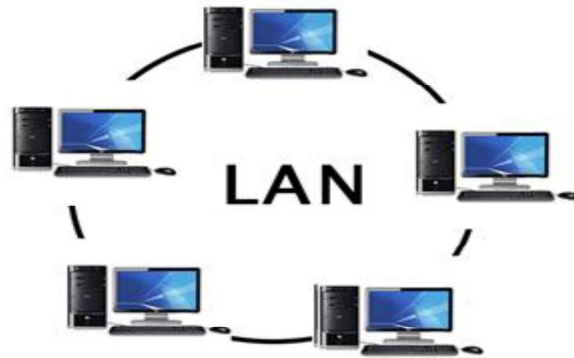


Fig. 3.2 Local Area Network

PAN (Personal Area Network)

Personal Area Network is a network arranged within an individual person, typically within a range of 10 meters.

- Personal Area Network is used for connecting the computer devices of personal use is known as Personal Area Network.
- Personal Area Network covers an area of **30 feet**.
- Personal computer devices that are used to develop the personal area network are the laptop, mobile phones, media player and play stations as shown in Figure 3.3.



Fig. 3.3 Personal Area Network

MAN (Metropolitan Area Network)

A metropolitan area network is a network that covers a larger geographic area by interconnecting a different LAN to form a larger network (see Figure 3.4).

- Government agencies use MAN to connect to the citizens and private industries.
- In MAN, various LANs are connected to each other through a telephone exchange line.
- The most widely used protocols in MAN are RS-232, Frame Relay, ATM, ISDN, OC-3, ADSL, etc.
- It has a higher range than Local Area Network (LAN).

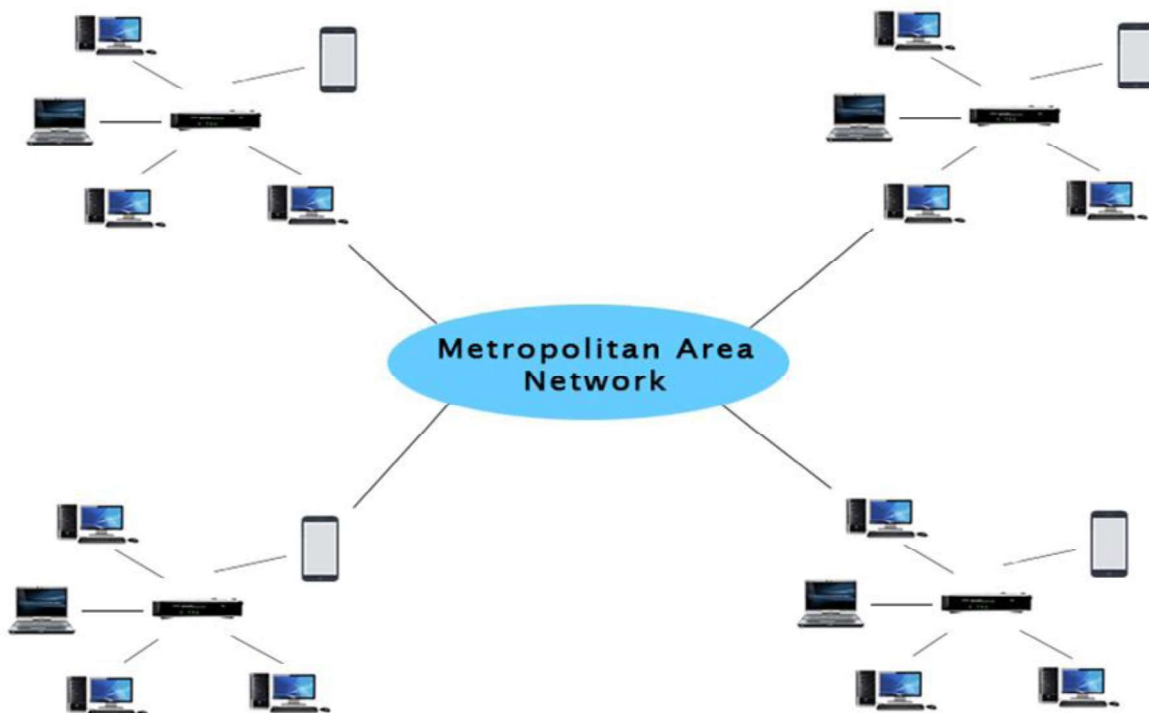


Fig. 3.4 Metropolitan Area Network

WAN (Wide Area Network)

A Wide Area Network (Figure 3.5) is a network that extends over a large geographical area such as states or countries.

- A Wide Area Network is quite bigger network than the LAN.

- A Wide Area Network is not limited to a single location, but it spans over a large geographical area through a telephone line, fibre optic cable or satellite links.
- The internet is one of the biggest WAN in the world.
- A Wide Area Network is widely used in the field of business, government, and education.

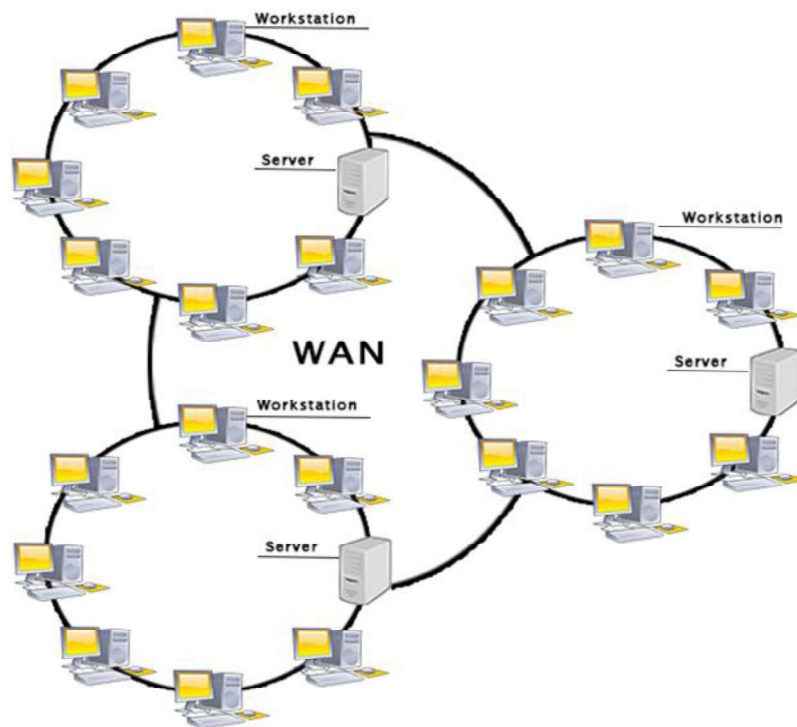


Fig. 3.5 Wide Area Network

3.1.2 Importance of Computer Networking

In the world of Information Technology (IT), information is a building block for effective communication. Communication is medium that helps us to drive our day to day professional and personal operations. Computer networks helps us to achieve the following:

- Provides best way of business communication.
- Streamline communication.
- Cost-effective resource sharing.
- Improving storage efficiency and volume.
- Cut costs on software.

- Cut costs on hardware.
- Utilizes Centralized Database.
- Increase in efficiency.
- Optimize convenience and flexibility.
- Allows file sharing.
- Sharing of peripherals and internet access.
- Network gaming.
- Voice over IP (VoIP).
- Media Center Server.
- Centralize network administration, meaning less IT support.
- Flexibility.
- Allowing information sharing.
- Supporting distributed processing.
- User communication.
- Overcoming geographic separation.

In general, we can see “**Communication**” and “**Sharing**”, that is all networking is about in simple terms. It enables everything that we do today. In the current scenario, the business would not work without computer networking and our personal and social life will also feel like it has been stopped.

3.1.3 Basic Computer Network Components

Computer networks share common devices, functions, and features including servers, clients, transmission media, shared data, shared printers and other hardware and software resources, network interface card (NIC), local operating system (LOS), and the network operating system (NOS).

Servers - Servers are computers that hold shared files, programs, and the network operating system. Servers provide access to network resources to all the users of the network. There are many different kinds of servers, and one server can provide several functions. For example, there are file servers, print servers, mail servers, communication

servers, database servers, fax servers and web servers, to name a few. Sometimes it is also called host computer, servers are powerful computer that store data or application and connect to resources that are shared by the user of a network.

Clients - Clients are computers that access and use the network and shared network resources. Client computers are basically the customers (users) of the network, as they request and receive services from the servers. These days, it is typical for a client to be a personal computer that the users also use for their own non-network applications.

Transmission Media - Transmission media are the facilities used to interconnect computers in a network, such as twisted-pair wire, coaxial cable, and optical fiber cable. Transmission media are sometimes called transmission medium channels, links or lines.

Shared data - Shared data are data that file servers provide to clients such as data files, printer access programs and e-mail.

Shared printers and other peripherals - Shared printers and peripherals are hardware resources provided to the users of the network by servers. Resources provided include data files, printers, software, or any other items used by clients on the network.

Network Interface Card - Each computer in a network has a special expansion card called a network interface card (NIC). The NIC prepares (formats) and sends data, receives data, and controls data flow between the computer and the network. On the transmit side, the NIC passes frames of data on to the physical layer, which transmits the data to the physical link. On the receiver's side, the NIC processes bits received from the physical layer and processes the message based on its contents.

Local Operating System - A local operating system allows personal computers to access files, print to a local printer, and use one or more disk and CD drives that are located on the computer. Examples are MS-DOS, Unix, Linux, Windows 2000, Windows 98, Windows XP etc. The network operating system is the software of the network. It serves a similar purpose that the OS serves in a stand-alone computer

Network Operating System - The network operating system is a program that runs on computers and servers that allows the computers to communicate over the network.

Hub - Hub is a device that splits a network connection into multiple computers. It is like a distribution center. When a computer requests information from a network or a specific computer, it sends the request to the hub through a cable. The hub will receive the request and transmit it to the entire network. Each computer in the network should then figure out whether the broadcast data is for them or not.

Switch - Switch is a telecommunication device grouped as one of computer network components. Switch is like a Hub but built in with advanced features. It uses physical device addresses in each incoming messages so that it can deliver the message to the right destination or port.

Unlike a hub, switch doesn't broadcast the received message to entire network, rather before sending it checks to which system or port should the message be sent. In other words, switch connects the source and destination directly which increases the speed of the network.

Router - The device used to connect a LAN with an internet connection is called Router. When you have two distinct networks (LANs) or want to share a single internet connection to multiple computers, we use a Router.

LAN Cable - A local area network cable is also known as data cable or Ethernet cable which is a wired cable used to connect a device to the internet or to other devices like computer, printers, etc.

3.2 Network Cable Types and Specifications

To connect two or more computers or networking devices in a network, network cables are used. There are three types of network cables; coaxial, twisted-pair, and fiber-optic.

Coaxial cable

This cable contains a conductor, insulator, braiding, and sheath. The sheath covers the braiding, the braiding covers the insulation, and the insulation covers the conductor. The following image (Figure 3.6) shows these components.

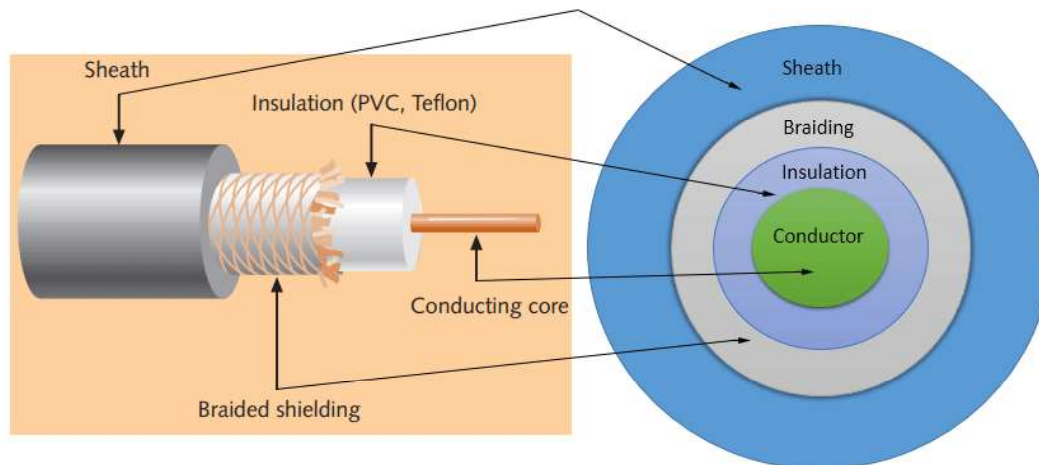


Fig. 3.6 Coaxial Cable

Twisted-pair Cables

The twisted-pair cable was primarily developed for computer networks. This cable is also known as **Ethernet cable**. Almost all modern LAN computer networks use this cable.

This cable consists of color-coded pairs of insulated copper wires. Every two wires are twisted around each other to form pair. Usually, there are four pairs. Each pair has one solid color and one stripped color wire. Solid colors are blue, brown, green, and orange. In stripped color, the solid color is mixed with the white color. Based on how pairs are stripped in the plastic sheath, there are two types of twisted-pair cable; UTP and STP.

In the **UTP (Unshielded twisted-pair) cable**, all pairs are wrapped in a single plastic sheath.

In the **STP (Shielded twisted-pair) cable**, each pair is wrapped with an additional metal shield, then all pairs are wrapped in a single outer plastic sheath.

The following image (Figure 3.7) shows both types of twisted-pair cables.

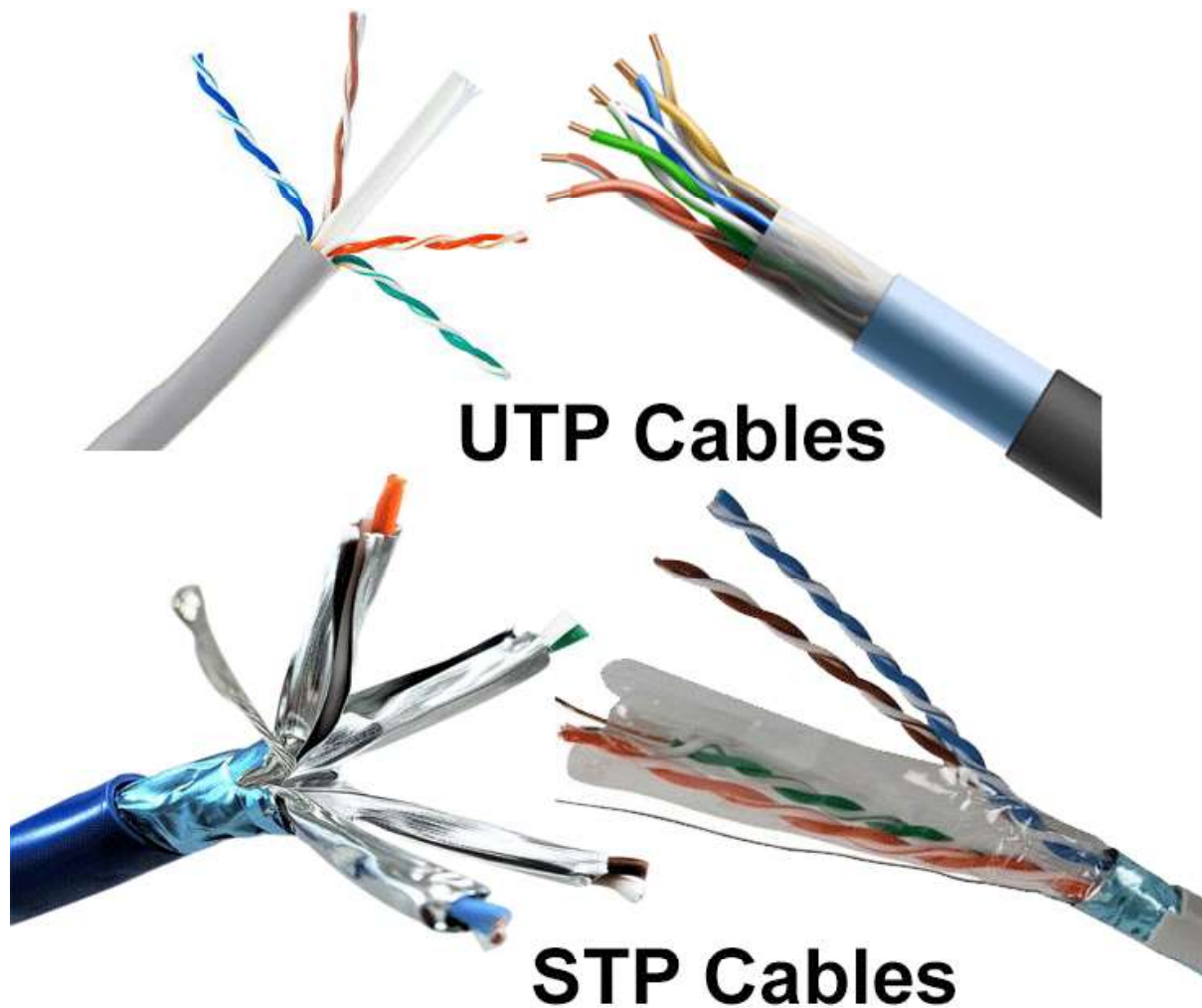


Fig. 3.7 Twisted-pair Cables

Fiber Optic Cable

This cable consists of a core, cladding, buffer, and jacket. The core is made from thin strands of glass or plastic that can carry data over a long distance. The core is wrapped in the cladding; the cladding is wrapped in the buffer, and the buffer is wrapped in the jacket. Fiber optic cable is completely immune to EMI and RFI. This cable can transmit data over a long distance at the highest speed. It can transmit data up to 40 kilometres at the speed of 100Gbps. Fiber optic uses light to send data. It reflects light from one endpoint to another. Based on how many beams of light are transmitted at a given time, there are two types of fiber optical cable; SMF and MMF as shown in Figure 3.8.

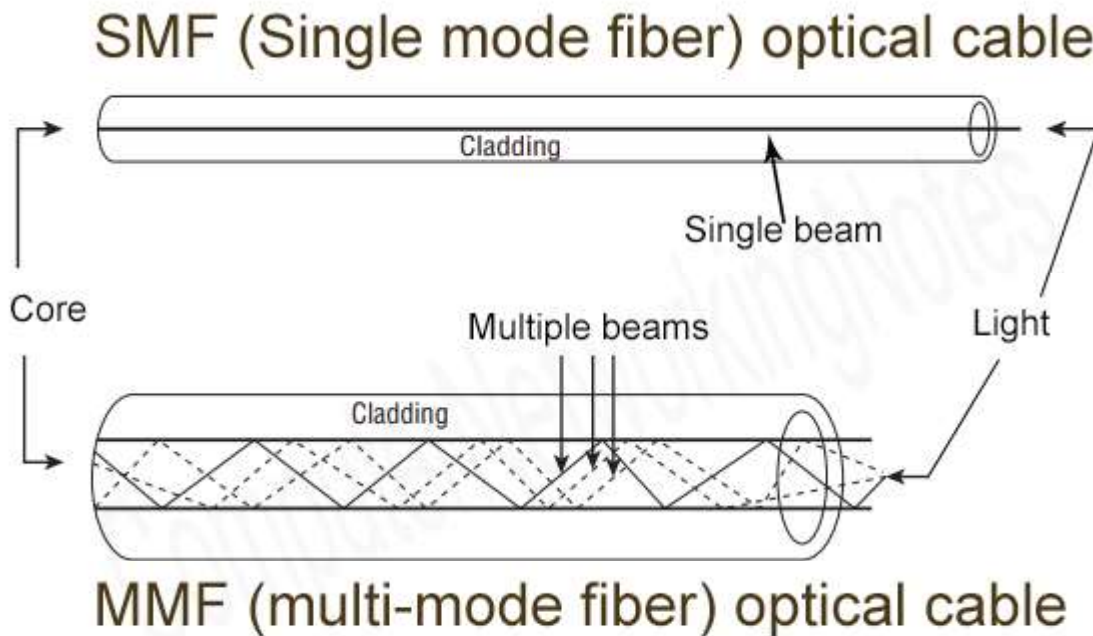


Fig. 3.8 Fiber Optic Cable

What is an Ethernet cable?

An Ethernet cable is a network cable used for high-speed wired network connections between two devices. This network cable is made of a four-pair cable, which consists of twisted pair conductors. It is used for data transmission at both ends of the cable, which is called RJ45 connector.

The Ethernet cables are categorized as CAT5, CAT5e, CAT6, and UTP cable. CAT5 cable can support a 10/100 Mbps Ethernet network while Cat5e and Cat 6 cable to support Ethernet network running at 10/100/1000 Mbps.

What is straight through Cable?

Straight-through cable is a type of CAT5 with RJ-45 connectors at each end, and each has the same pin out as shown in Figure 3.9. It is in accordance with either the T568A or T568B standards. It uses the same color code throughout the LAN for consistency. This type of twisted-pair cable is used in LAN to connect a computer or a network hub such as a router. It is one of the most common types of network cable.

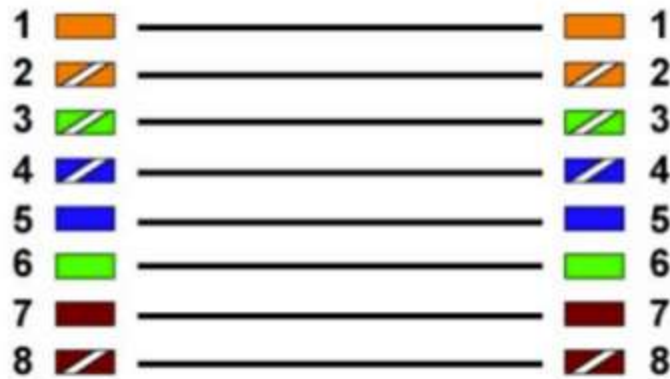


Fig. 3.9 Straight Through Cable Connection

What Is Crossover Cable?

A Crossover cable is a type of CAT5 where one end is T568A configuration and the other end as T568B Configuration. In this type of cable connection, Pin 1 is crossed with Pin 3, and Pin 2 is crossed with Pin 6 as shown in Figure 3.10. Crossover cable is used to connect two or more computing devices. The internal wiring of crossover cables reverses the transmission and receive signals. It is widely used to connect two devices of the same type: e.g., two computers or two switches to each other.

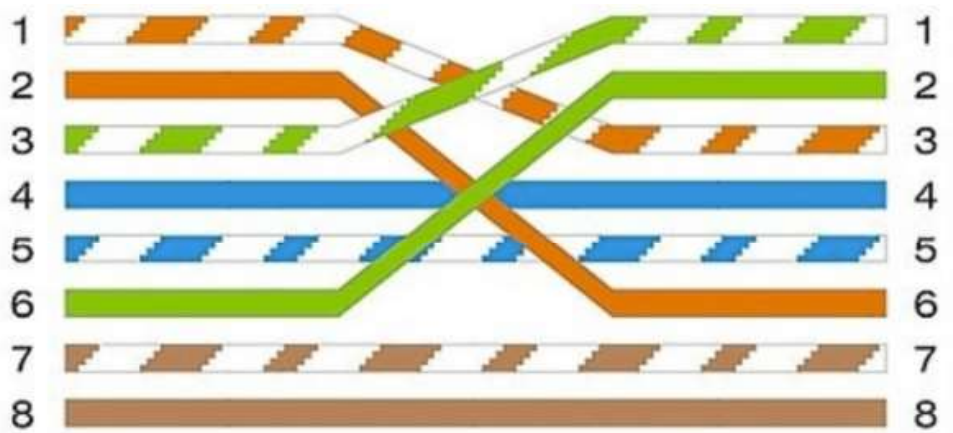


Fig. 3.10 Cross Over Cable Connection

RJ 45 and RJ 11:

RJ is the acronym for Registered Jack and refers to the industry standard character that cable connectors usually start with. RJ45 and RJ11 are the 2 most command connector standards used universally. The main difference between these two is in where they are

actually used. RJ45 jacks are used in networking (Ethernet LAN), where you connect computers or other network elements to each other. RJ11 is the cable connector that is being used in telephone sets.

Using Cable Tester to Check Connection

Figure 3.11 shows a cable tester. It is used to check connection between two ends of a cable. Follow the steps given below to check an Ethernet wire using cable tester.



Fig. 3.11 Cable Tester

1. Plug one end of the cable into the transmit jack. The transmit jack on the tester maybe be labelled "TX".
2. Plug the other end of the cable into the receiver jack. The receiver jack may be labelled "RX" on the device. Some testers have a detachable receiver you can use to test the cable across different rooms.
3. Check the lights on the tester. Most testers will have 2 sets of 8 LED lights that correspond to the 8 pins on the transmitting and receiving end of the Ethernet cable. There will also be a G light for the ground. It will test each pin one at a time. If all 8 pins light up on both ends, the cable is good. If any of the lights do

not light up on either end, this indicates a short in the cable. Don't worry if the G light doesn't light up. If the lights on either end flash out of order, this indicates that you are testing a crossover cable. As long as all 8 lights flash, the cable is good.

Fixing a RJ45 Connector

1. Strip the cable back 1 inch (25 mm) from the end.
2. Untwist and straighten the wires inside of the cable.
3. Arrange the wires into the right order.
4. Cut the wires into an even line $\frac{1}{2}$ inch (13 mm) from sheathing.
5. Insert the wires into the RJ-45 connector.
6. Stick the connector into the crimping part of the tool and squeeze twice.
7. Remove the cable from the tool and check that all of the pins are down.

Wireless vs Wired Network

A wireless network allows devices to stay connected to the network but roam untethered to any wires. Access points amplify Wi-Fi signals, so a device can be far from a router but still be connected to the network. When you connect to a Wi-Fi hotspot at a cafe, a hotel, an airport lounge, or another public place, you're connecting to that business's wireless network.

A wired network uses cables to connect devices, such as laptop or desktop computers, to the Internet or another network. A wired network has some disadvantages when compared to a wireless network. The biggest disadvantage is that your device is tethered to a router. The most common wired networks use cables connected at one end to an Ethernet port on the network router and at the other end to a computer or other device.

Bit Rate:

The speed of the data is expressed in bits per second (bits/s or bps). The data rate R is a function of the duration of the bit. Rate is also called channel capacity C .

Teacher Notes:

- Give a demonstration of speedtest.net to students.

Baud Rate

Baud rate refers to the number of signal or symbol changes that occur per second. A symbol is one of several voltage, frequency, or phase changes. NRZ (Non Return to Zero) binary has two symbols, one for each bit 0 or 1, that represent voltage levels. In this case, the baud or symbol rate is the same as the bit rate. However, it's possible to have more than two symbols per transmission interval, whereby each symbol represents multiple bits. With more than two symbols, data is transmitted using modulation techniques. By using multiple symbols, multiple bits can be transmitted per symbol. For example, if the symbol rate is 4800 baud and each symbol represents two bits, that translates into an overall bit rate of 9600 bits/s. Normally the number of symbols is some power of two. If N is the number of bits per symbol, then the number of required symbols is $S = 2^N$.

Attenuation

Attenuation in computer networking is the loss of communication signal strength that is measured in decibels (dB). As the rate of attenuation increases, the transmission, such as an email a user is trying to send or a phone call, becomes more distorted. Attenuation occurs on computer networks because of:

- Range – over longer distances both wired and wireless transmissions gradually dissipate in strength
- Interference – radio interference or physical obstructions, such as walls, dampen communication signals on wireless networks
- Wire size – thinner wires suffer from more attenuation than thicker wires on wired networks

3.3 Network Devices

1. Repeater – A repeater operates at the physical layer. Its job is to regenerate the signal over the same network before the signal becomes too weak or corrupted so as to extend the length to which the signal can be transmitted over the same network. An important point to be noted about repeaters is that they do not amplify the signal. When the signal becomes weak, they copy the signal bit by bit and regenerate it at the original strength. It is a 2-port device.

2. Hub – A hub is basically a multiport repeater. A hub connects multiple wires coming from different branches, for example, the connector in star topology which connects different stations. Hubs cannot filter data, so data packets are sent to all connected devices. In other words, the collision domain of all hosts connected through Hub remains one. Also, they do not have the intelligence to find out the best path for data packets which leads to inefficiencies and wastage.

3. Bridge – A bridge operates at the data link layer. A bridge is a repeater, with add on the functionality of filtering content by reading the MAC addresses of source and destination. It is also used for interconnecting two LANs working on the same protocol. It has a single input and single output port, thus making it a 2-port device.

4. Switch – A switch is a multiport bridge with a buffer and a design that can boost its efficiency (a large number of ports imply less traffic) and performance. A switch is a data link layer device. The switch can perform error checking before forwarding data, which makes it very efficient as it does not forward packets that have errors and forward good packets selectively to the correct port only.

5. Routers – A router is a device like a switch that routes data packets based on their IP addresses. The router is mainly a Network Layer device. Routers normally connect LANs and WANs together and have a dynamically updating routing table based on which they make decisions on routing the data packets. Router divide broadcast domains of hosts connected through it.

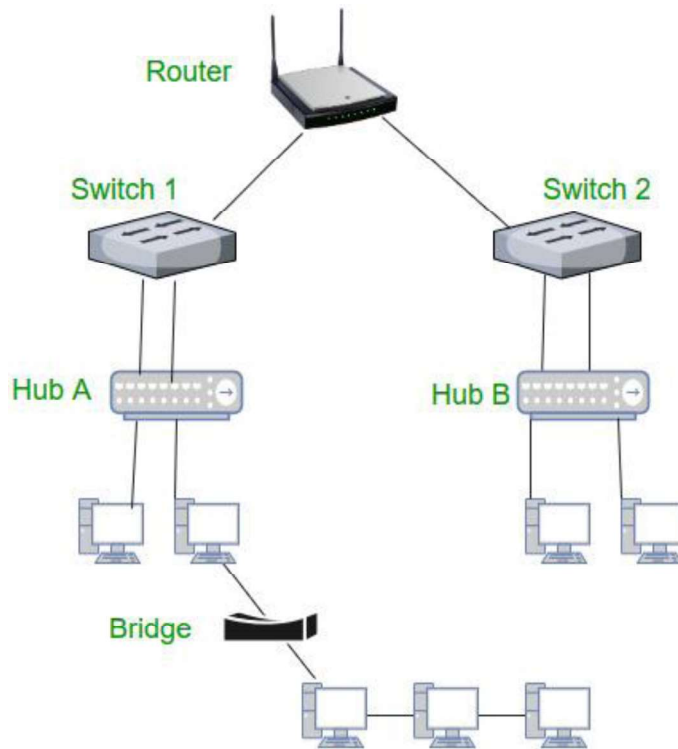


Fig. 3.12 Network Devices

6. Gateway – A gateway, as the name suggests, is a passage to connect two networks together that may work upon different networking models. They basically work as the messenger agents that take data from one system, interpret it, and transfer it to another system. Gateways are also called protocol converters and can operate at any network layer. Gateways are generally more complex than switches or routers. Gateway is also called a protocol converter.

7. NIC – NIC (Network Interface Card) is a network adapter that is used to connect the computer to the network. It is installed in the computer to establish a LAN. It has a unique id that is written on the chip, and it has a connector to connect the cable to it. The cable acts as an interface between the computer and router or modem. NIC card is a layer 2 device which means that it works on both physical and data link layer of the network model.

3.3.1 Router Interfaces

Routers can have many different types of connectors; from Ethernet, Fast Ethernet, and Token Ring to Serial and ISDN ports. Some of the available configurable items are logical addresses (IP, IPX), media types, bandwidth, and administrative commands. Interfaces are configured in interface mode which is available from global configuration mode after logging in. Generally, three types of interfaces are provided to manage a Network switch:

Simple Network Management Protocol:

Also known as SNMP, it is a protocol that facilitates the exchange of management information between network devices. **SNMP** queries can determine the health of a network or the status of a particular device.

Command Line Interface:

Command Line Interface (CLI) is an exhaustive set of commands that can be accessed via serial console, Telnet, and Secure Shell.

The Web Interface:

User can monitor and configure the switch from a **browser**, by entering a URL.

3.3.2 Wireless Devices and their Roles

There are three major “modes” a Wi-Fi device can use. These modes define the role a Wi-Fi device has in the network, and networks must be built out of combinations of devices operating in these different modes. How the devices are configured depends on the types of connections you want to use between parts of the network.

The three wireless roles are

1. Wireless Clients (Station)

Devices such as computers, tablets, and phones are common clients on a network. When you are accessing a wireless hotspot, or the router in your home or office, your device is the client. This client mode is also known as “station mode” as well.

2. Access Points (Master)

Most wireless networks are made using Access Points - devices that host and control the wireless connection for laptops, tablets, or smart phones. If you use Wi-Fi in your home or office, it is most likely through an Access Point. When a router is set up as an AP, it is said to be in “Master” or “Infrastructure” mode.

An AP is sometimes a stand-alone device that bridges between a wireless and wired (Ethernet) network, or is part of a router. APs can cover a range of areas with a wireless signal, depending on the power of the device and the type of antenna. There are also some APs that are weatherproof, designed to be mounted outdoors.

3. Ad-Hoc Node (Mesh)

Some wireless devices (laptops, smart phones, or wireless routers) support a mode called Ad-Hoc. This allows those devices to connect together directly, without an Access Point in-between controlling the connection. This forms a different type of network - in Ad-Hoc mode, all devices are responsible for sending and receiving messages to the other devices - without anything else in between. In an Ad-Hoc network, every device must be in this role, and using the same configuration to participate.

3.4 Network Topologies

A Network Topology is the arrangement with which computer systems or network devices are connected to each other. Topologies may define both physical and logical aspect of the network. Both logical and physical topologies could be same or different in a same network.

Point-to-Point

Point-to-point networks contains exactly two hosts such as computer, switches or routers, servers connected back-to-back using a single piece of cable. Figure 3.13 shows a Point-to-point network. Often, the receiving end of one host is connected to sending end of the other and vice-versa.

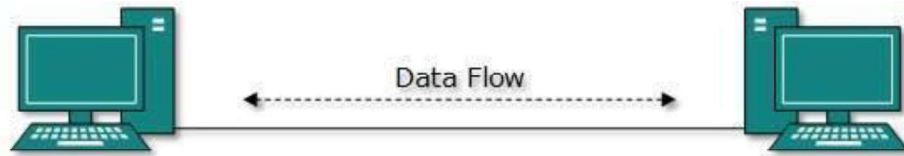


Fig. 3.13 Point-to-point Networks

If the hosts are connected point-to-point logically, then may have multiple intermediate devices. But the end hosts are unaware of underlying network and see each other as if they are connected directly.

Bus Topology

In case of Bus topology, all devices share single communication line or cable. Bus topology may have problem while multiple hosts sending data at the same time. It is one of the simple forms of networking where a failure of a device does not affect the other devices. But failure of the shared communication line can make all other devices stop functioning.

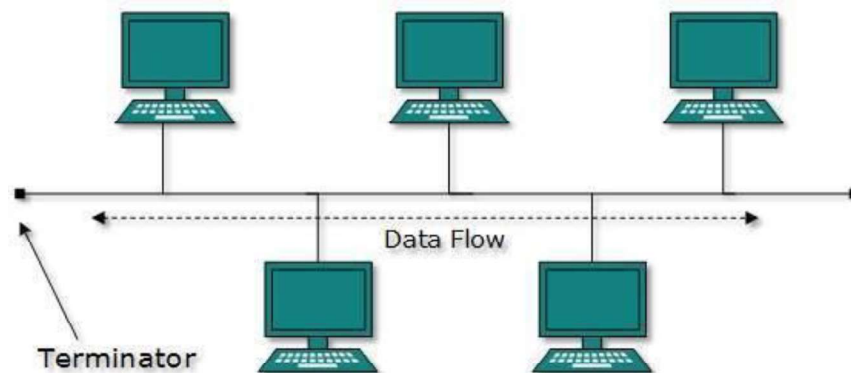


Fig. 3.14 Bus Topology

Both ends of the shared channel have line terminator as shown in Figure 3.14. The data is sent in only one direction and as soon as it reaches the extreme end, the terminator removes the data from the line.

Star Topology

All hosts in Star topology are connected to a central device, known as hub device, using a point-to-point connection. That is, there exists a point-to-point connection between hosts and hub. The hub device can be any of the following:

- Layer-1 device such as hub or repeater
- Layer-2 device such as switch or bridge
- Layer-3 device such as router or gateway

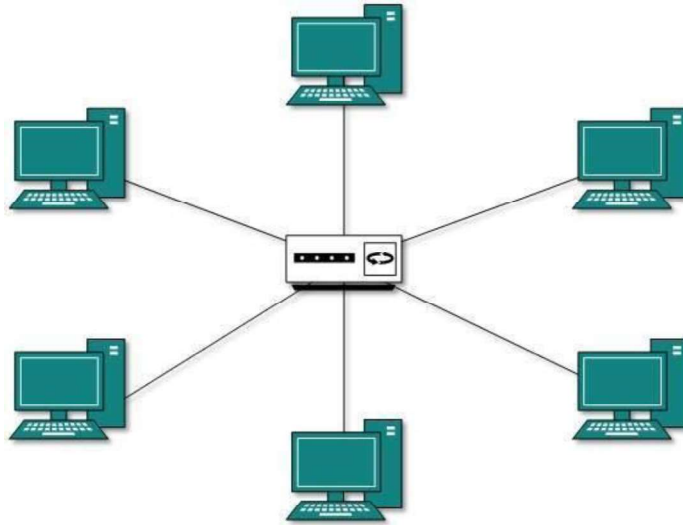


Fig. 3.15 Star Topology

As in Bus topology, if the single point of connection fails, connectivity of all hosts to all other hosts fails. Every communication between hosts, takes place through only the hub. Star topology is not expensive as to connect one more host, only one cable is required and configuration is simple.

Ring Topology

In ring topology, each host machine connects to exactly two other machines, creating a circular network structure. When one host tries to communicate or send message to a host which is not adjacent to it, the data travels through all intermediate hosts. To connect one more host in the existing structure, the administrator may need only one more extra cable. Ring topology is shown in Figure 3.16.

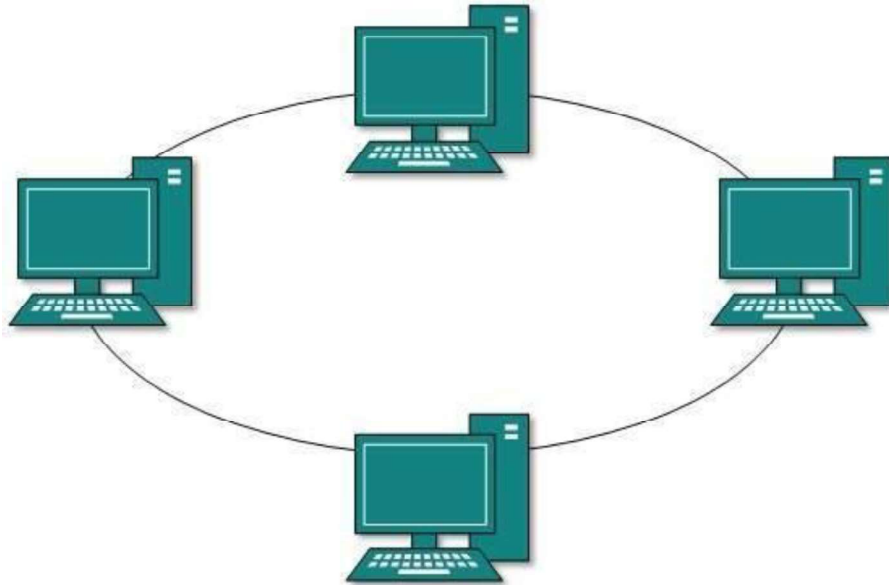


Fig. 3.16 Ring Topology

Failure of any host results in failure of the whole ring. Thus, every connection in the ring is a point of failure. There are methods which employ one more backup ring.

Mesh Topology

In this type of topology, a host is connected to one or multiple hosts. This topology has hosts in point-to-point connection with every other host or may also have hosts which are in point-to-point connection to few hosts only.

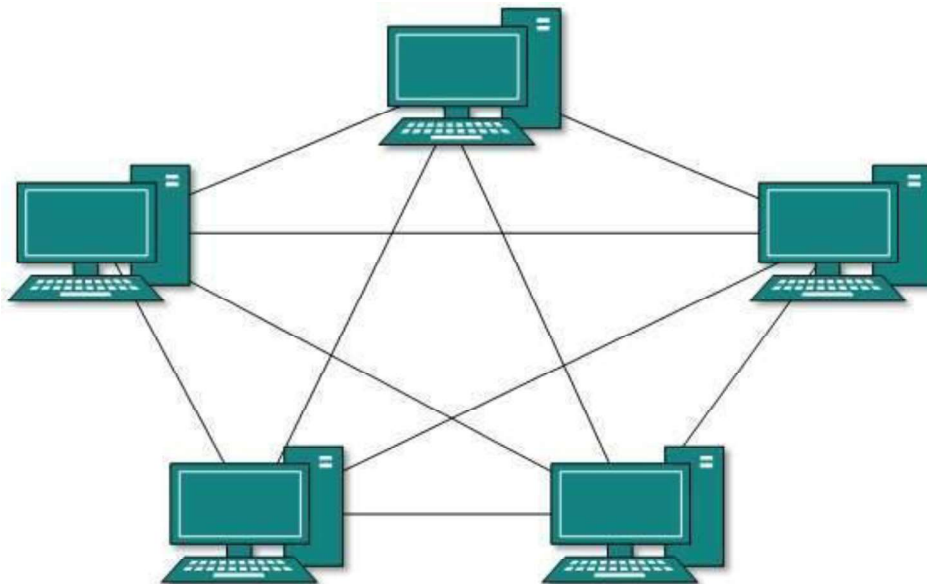


Fig. 3.17 Mesh Topology

Active Network Components vs Passive Network Components

Active Networking Hardware comprises all parts of a computer network which in some way actively work with the signals in the network (amplify them, modify them, evaluate them, etc.). Active Networking Hardware are usually specific devices located in the hubs of the network. Active networking hardware includes:

- Switch
- Repeater
- Hub
- Bridge
- Router
- Print Server
- Access point (AP)

Passive Networking Hardware are those parts of computer network that are involved in data transmission in the network, but they don't change or affect the data. The passive network hardware includes:

- Cables (fiber optic cable, coaxial cables)
- Connectors
- Switchboards

3.5 Configuring a Router

Network configuration is the process of setting a network's controls, flow and operation to support the network communication of an organization and/or network owner. This broad term incorporates multiple configuration and setup processes on network hardware, software and other supporting devices and components.

Before configuring, connect a computer to TP-Link router by cable or through wireless network as shown in Figure 3.18. It is recommended to configure it by a wired connection.



Fig. 3.18 Setting up a Router

Log into the router's web-based utility and follow the following steps:

Step 1

Select **Wireless->Wireless Settings** on the left side menu to open the wireless setting page as shown in Figure 3.19.

Wireless Network Name (also called SSID for some models): Create a new name for your wireless network. If you want to use the default TP-Link_***** wireless name, you can also leave it here as default value.

Region: Select your current location.

"Enable Wireless Router Radio" and **"Enable SSID Broadcast"** should be both ticked.

Please do not change other settings on this page if not necessary.

Wireless Settings

Wireless Network Name: TP-LINK_37DE50 (Also called the SSID)

Region: United States

Warning: Ensure you select a correct country to conform local law. Incorrect settings may cause interference.

Mode: 11bgn mixed

Channel Width: Auto

Channel: Auto

Max Tx Rate: 300Mbps

Please use the WiFi button on this device to enable/disable radio.

☒ Enable Wireless Router Radio

☒ Enable SSID Broadcast

☐ Enable WDS Bridging

Save

Fig. 3.19 Wireless Settings of a Router

Step 2

Select **Wireless->Wireless Security** on the left side menu as shown in Figure 3.20. It's recommended to use **WPA/WPA2 Personal (Recommended)**, some models will name that as **WPA-PSK/WPA2-PSK** instead.

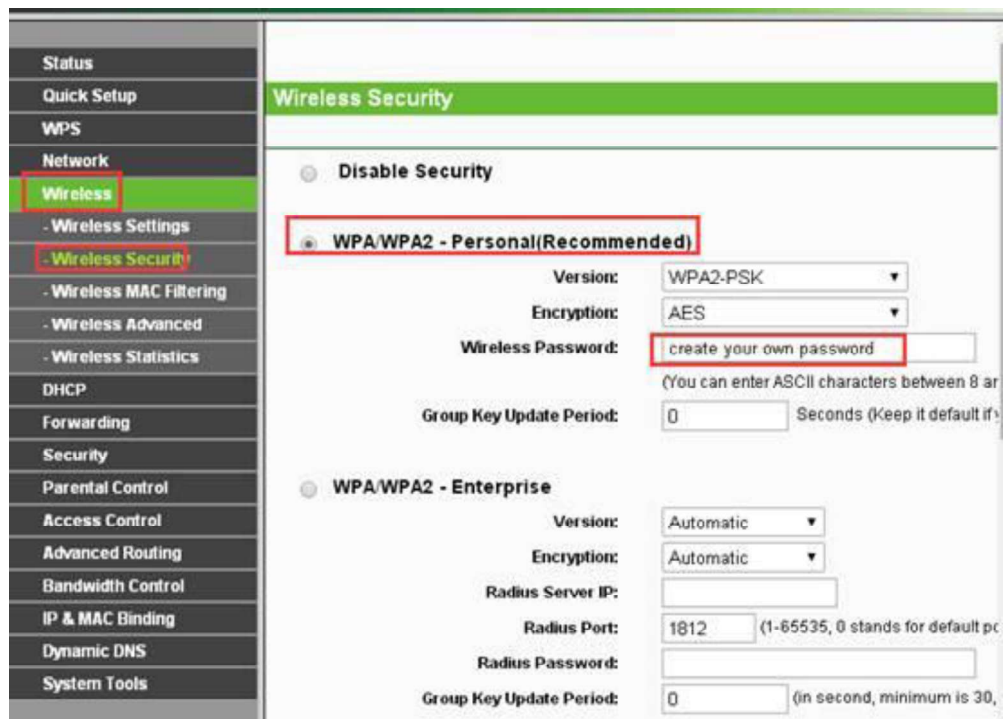


Fig. 3.20 Wireless Security Settings of a Router

Encryption: AES

Wireless Password (also called PSK Password): make up your Wi-Fi password, also called Wireless Network Key.

Setting IPv6

Step 1: Log in to the router's web-based management page.

Step 2: Go to **IPv6 Support** at the left-hand side menu as shown in Figure 3.21. In the drop-down menu of **IPv6 Support**, please choose **IPv6 Setup**.

Step 3: Please make sure that **Enable IPv6** has been checked. To Configure **WAN Connection Type**, if you are not sure what the connection type is, please contact your IPv6 provider. Here, take **PPPoEv6** as an example. After the **PPPoEv6** is selected, please input the **User name** and **Password** provided by the IPv6 Provider. For the **Get IPv6 Address Way**, please choose **Get IPv6 prefix delegation**. For the **IPV6 Address Assign Type**, please choose **SLAAC**.

Status

Quick Setup

Network

Dual Band Selection

Wireless 2.4GHz

Wireless 5GHz

Guest Network

DHCP

USB Settings

NAT

Forwarding

Security

Parental Control

Access Control

Advanced Routing

Bandwidth Control

IP & MAC Binding

Dynamic DNS

IPv6 Support

- IPv6 Status

IPv6 Setup

System Tools

Logout

WAN Setup

Enable IPv6:

☒

WAN Connection Type:

PPPoEv6

▼

User Name:

Password:

Confirm Password:

Get IPv6 Address Way:

Get IPv6 prefix delegation

▼

IPv6 Address:

::

Connect

Disconnect

Disconnected!

LAN Setup

IPv6 Address Assign Type:

SLAAC

▼

IPv6 Address Prefix:

::

/64

LAN IPv6 Address:

Save

Advanced

Fig. 3.21 IPv6 Settings of a Router

IP settings of a Network

If your ISP provides the DHCP service, please select *Dynamic IP* (as shown in Figure 3.22), and the router will automatically get IP parameters from your ISP.

The screenshot shows the 'WAN Settings' interface. At the top, 'Connection Type' is set to 'Dynamic IP' with a 'Detect' button. Below this, 'IP Address', 'Subnet Mask', and 'Gateway' are all set to '0.0.0.0'. There are 'Renew' and 'Release' buttons. A horizontal separator line is followed by 'MTU(Bytes)' set to '1500' with a note '(1500 as default, do not change unless necessary)' and a 'Hide' button. Below the separator, 'Enable IGMP Proxy' is checked. 'IGMP Version' has radio buttons for 'v2' and 'v3', with 'v3' selected. 'Get IP with Unicast' is unchecked with a note '(It is usually not required)'. 'Set DNS server manually' is also unchecked. At the bottom, there is a 'Host Name' field and a 'Save' button.

Fig. 3.22 IPv6 Settings of a Router

Static IP

If your ISP provides a static or fixed IP address, subnet mask, default gateway and DNS setting, please select Static IP in Figure 3.22 and fill the following fields:

- IP Address** - Enter the IP address in dotted-decimal notation provided by your ISP.
- Subnet Mask** - Enter the subnet mask in dotted-decimal notation provided by your ISP.
Normally 255.255.255.0 is used as the subnet mask.
- Gateway** - Enter the gateway IP address in dotted-decimal notation provided by your ISP.
- Primary/Secondary DNS Server** - (Optional) Enter one or two DNS addresses in dotted-decimal notation provided by your ISP.
- MTU (Bytes)** - The normal MTU (Maximum Transmission Unit) value for most Ethernet networks is 1500 Bytes. It is not recommended that you change the default MTU size unless required by your ISP.
- Enable IGMP Proxy** - IGMP (Internet Group Management Protocol) is used to manage multicasting on TCP/IP networks. Some ISPs use IGMP to perform remote

configuration for client devices, such as the modem router. The default value is enabled, and if you are not sure, please contact your ISP or just leave it.

3.6 OSI Model

A typical OSI layered model is shown in Figure 3.23. We'll describe OSI layers “top down” from the application layer that directly serves the end user, down to the physical layer.

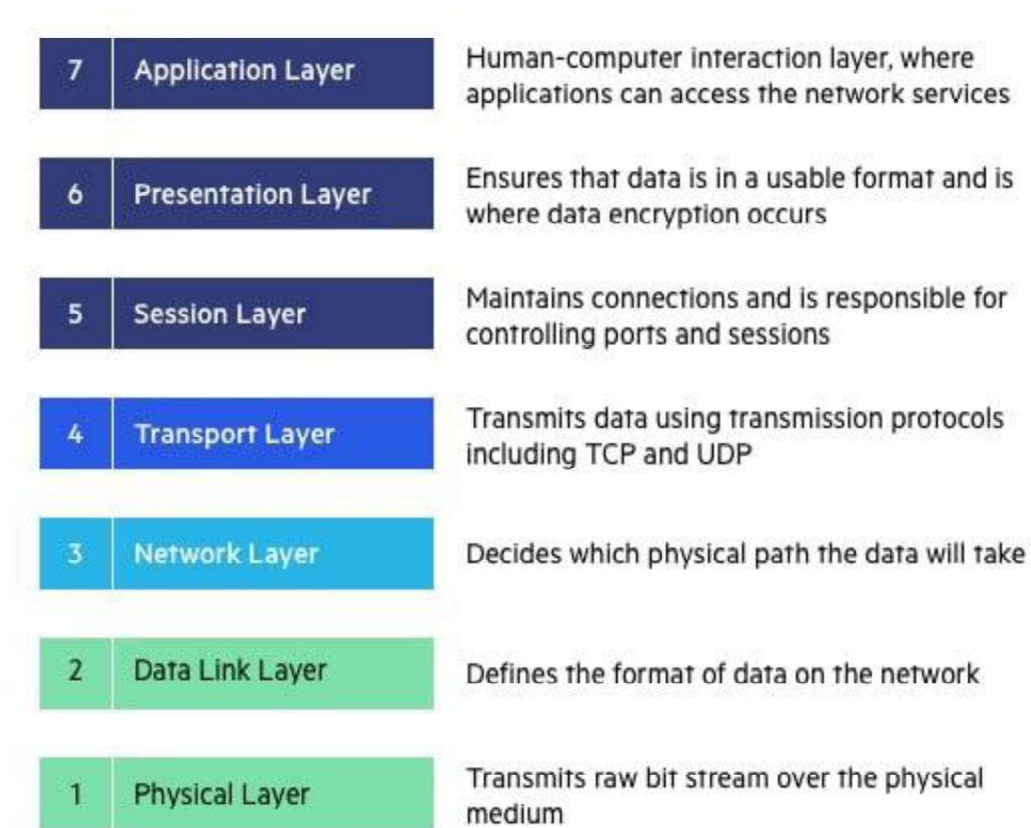


Fig. 3.23 OSI Layered Model

Application Layer: The application layer is used by end-user software such as web browsers and email clients. It provides protocols that allow software to send and receive information and present meaningful data to users. A few examples of application layer protocols are the Hypertext Transfer Protocol (HTTP), File Transfer Protocol (FTP), Post Office Protocol (POP), Simple Mail Transfer Protocol (SMTP), and Domain Name System (DNS).

Presentation Layer: The presentation layer prepares data for the application layer. It defines how two devices should encode, encrypt, and compress data so it is received correctly on the other end. The presentation layer takes any data transmitted by the application layer and prepares it for transmission over the session layer.

Session Layer: The session layer creates communication channels, called sessions, between devices. It is responsible for opening sessions, ensuring they remain open and functional while data is being transferred, and closing them when communication ends. The session layer can also set checkpoints during a data transfer—if the session is interrupted, devices can resume data transfer from the last checkpoint.

Transport Layer: The transport layer takes data transferred in the session layer and breaks it into “segments” on the transmitting end. It is responsible for reassembling the segments on the receiving end, turning it back into data that can be used by the session layer. The transport layer carries out flow control, sending data at a rate that matches the connection speed of the receiving device, and error control, checking if data was received incorrectly and if not, requesting it again.

Network Layer: The network layer has two main functions. One is breaking up segments into network packets, and reassembling the packets on the receiving end. The other is routing packets by discovering the best path across a physical network. The network layer uses network addresses (typically Internet Protocol addresses) to route packets to a destination node.

Data Link Layer: The data link layer establishes and terminates a connection between two physically-connected nodes on a network. It breaks up packets into frames and sends them from source to destination. This layer is composed of two parts—Logical Link Control (LLC), which identifies network protocols, performs error checking and synchronizes frames, and Media Access Control (MAC) which uses MAC addresses to connect devices and define permissions to transmit and receive data.

Physical Layer: The physical layer is responsible for the physical cable or wireless connection between network nodes. It defines the connector, the electrical cable or

wireless technology connecting the devices, and is responsible for transmission of the raw data, which is simply a series of 0s and 1s, while taking care of bit rate control.

TCP/IP model

The TCP/IP model was developed prior to the OSI model. The TCP/IP model is not exactly similar to the OSI model.

- The TCP/IP model consists of five layers: the application layer, transport layer, network layer, data link layer and physical layer.
- The first four layers provide physical standards, network interface, internetworking, and transport functions that correspond to the first four layers of the OSI model and these four layers are represented in TCP/IP model by a single layer called the application layer.
- TCP/IP is a hierarchical protocol made up of interactive modules, and each of them provides specific functionality.

Here, hierarchical means that each upper-layer protocol is supported by two or more lower-level protocols.

OSI vs. TCP/IP Model

The Transfer Control Protocol/Internet Protocol (TCP/IP) is older than the OSI model and was created by the US Department of Defence (DoD). A key difference between the models is that TCP/IP is simpler, collapsing several OSI layers into one:

- OSI layers 5, 6, 7 are combined into one Application Layer in TCP/IP as shown in Figure 3.24.
- OSI layers 1, 2 are combined into one Network Access Layer in TCP/IP – however TCP/IP does not take responsibility for sequencing and acknowledgement functions, leaving these to the underlying transport layer.
- TCP/IP is a functional model designed to solve specific communication problems, and which is based on specific, standard protocols. OSI is a generic, protocol-independent model intended to describe all forms of network communication.

- In TCP/IP, most applications use all the layers, while in OSI simple applications do not use all seven layers. Only layers 1, 2 and 3 are mandatory to enable any data communication.

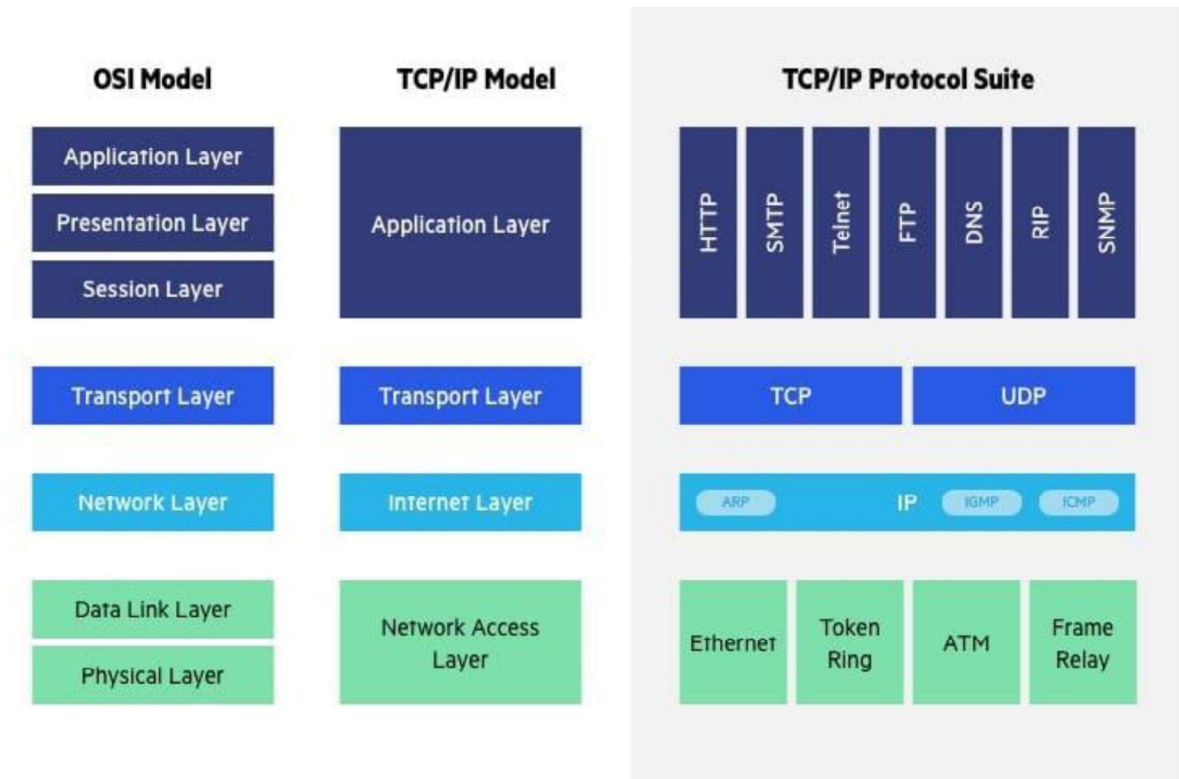


Fig. 3.24 OSI vs TCP/IP Model

What is IP?

An **IP (Internet Protocol) address** is a numerical label assigned to each device connected to a computer network that uses the IP protocol for communication. An IP address acts as an identifier for a specific device on a particular network. The IP address is also called an IP number or Internet address. IP address specifies the technical format of the addressing and packets scheme. Most networks combine IP with a TCP (Transmission Control Protocol). It also allows developing a virtual connection between a destination and a source.

IPv4

IPv4 is an IP version widely used to identify devices on a network using an addressing system. It was the first version of IP deployed for production in the ARPANET in 1983. It uses a 32-bit address scheme to store 2^{32} addresses which is more than 4 billion addresses. It is considered the primary Internet Protocol and carries 94% of Internet traffic.

IPv6

IPv6 is the most recent version of the Internet Protocol. This new IP address version is being deployed to fulfill the need for more Internet addresses. It was aimed to resolve issues that are associated with IPv4. With 128-bit address space, it allows 340 undecillion unique address space.

Static IP address:

A static IP address is simply an address that doesn't change. Once your device is assigned a static IP address, that number typically stays the same until the device is decommissioned or your network architecture changes. Static IP addresses generally are used by servers or other important equipment. Static IP addresses are assigned by Internet Service Providers (ISPs). Your ISP may or may not allocate you a static IP address depending on the nature of your service agreement.

Dynamic IP address

As the name suggests, dynamic IP addresses are subject to change, sometimes at a moment's notice. Dynamic addresses are assigned, as needed, by Dynamic Host Configuration Protocol (DHCP) servers. We use dynamic addresses because IPv4 doesn't provide enough static IP addresses to go around. So, for example, a hotel probably has a static IP address, but each individual device within its rooms would have a dynamic IP address. On the internet, your home or office may be assigned a dynamic IP address by your ISP's DHCP server.

MAC Address

The MAC address is a unique value associated with a network adapter. MAC addresses are also known as **hardware** addresses or **physical** addresses. They uniquely identify an adapter on a LAN. MAC addresses are 12-digit hexadecimal numbers (48 bits in length). By convention, MAC addresses are usually written in one of the following formats:

MM:MM:MM:SS:SS:SS

MMMM-MMSS-SSSS

Subnet mask:

A 32-bit combination used to describe which portion of an address refers to the subnet and which part refers to the host.

Classes of IP addresses

TCP/IP defines five classes of IP addresses: class A, B, C, D, and E. Each class has a range of valid IP addresses. The value of the first octet determines the class. IP addresses from the first three classes (A, B and C) can be used for host addresses. The other two classes are used for other purposes – class D for multicast and class E for experimental purposes.

3.7 Testing a Network

The basic ping test is one of the easiest things you can do to verify connectivity between your computer and a remote host. This test is easy to run from the command line.

Using ping test on Microsoft Windows

To use the ping program on Microsoft Windows, follow these steps:

1. Open a DOS command window. To do this, click **Start**, click **Run**, type *cmd*, and then press Enter.
2. At the command prompt, type the following command.
Replace *example.com* with the domain that you want to test:
Ping example.com
3. Interpret the output from ping:

1. If the remote host is active and configured to respond to ping requests, responses appear. For example, the following output shows ping responses from an A2 Hosting server:

```
C:\Documents and Settings\user>ping a2s78.a2hosting.com
```

- Pinging a2s78.a2hosting.com [216.119.143.98] with 32 bytes of data:
- Reply from 216.119.143.98: bytes=32 time=46ms TTL=54
- Reply from 216.119.143.98: bytes=32 time=45ms TTL=54
- Reply from 216.119.143.98: bytes=32 time=47ms TTL=54
- Ping statistics for 216.119.143.98:
- Packets: Sent = 3, Received = 3, Lost = 0 (0% loss),
- Approximate round-trip times in milli-seconds:

Minimum = 45ms, Maximum = 47ms, Average = 46ms

Alternatively, if the remote host is down, or not configured to respond to ping requests, you do not see any responses.

Using Router Interface

To check internet connectivity using router interface, follow the steps given below:

1. Open Router configuration page.
2. Click System Tools and Diagnostic.
3. Select one of the diagnostic tools: Ping.

Note: Ping troubleshoots connectivity, reachability, and name resolution to a given host or gateway.

4. Type the IP Address/Domain name of the PC/server you want to ping the IP Address/Domain Name, as shown in Figure 3.25.

5. Start Diagnose and analyze the results.

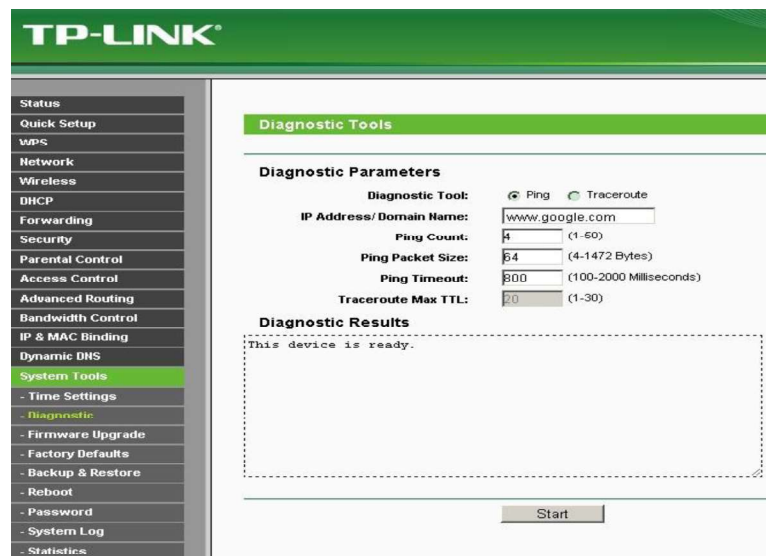


Fig. 3.25 Router Interface: Checking Internet

7	Application Layer	Human-computer interaction layer, where applications can access the network services
6	Presentation Layer	Ensures that data is in a usable format and is where data encryption occurs
5	Session Layer	Maintains connections and is responsible for controlling ports and sessions
4	Transport Layer	Transmits data using transmission protocols including TCP and UDP
3	Network Layer	Decides which physical path the data will take
2	Data Link Layer	Defines the format of data on the network
1	Physical Layer	Transmits raw bit stream over the physical medium

Key Points

- A computer network is a group of computers linked to each other that enables the computer to communicate with another computer and share their resources, data, and applications.
- Local Area Network is a group of computers connected to each other (as shown in Figure 3.2) in a small area such as building, office.
- Personal Area Network is a network arranged within an individual person, typically within a range of 10 meters.
- A metropolitan area network is a network that covers a larger geographic area by interconnecting a different LAN to form a larger network.
- A Wide Area Network is a network that extends over a large geographical area such as states or countries.
- Hub is a device that splits a network connection into multiple computers.
- Switch uses physical device addresses in each incoming messages so that it can deliver the message to the right destination or port.
- The device used to connect a LAN with an internet connection is called Router.
- An Ethernet cable is a network cable used for high-speed wired network connections between two devices.
- The speed of the data is expressed in bits per second (bits/s or bps).
- Baud rate refers to the number of signal or symbol changes that occur per second.
- Attenuation in computer networking is the loss of communication signal strength that is measured in decibels (dB).
- When you are accessing a wireless hotspot, or the router in your home or office, your device is the client.
- Access points are devices that host and control the wireless connection for laptops, tablets, or smart phones.
- Mac Address is the physical address of device.

Exercise

Select the most appropriate option

1. Which of the following is the hardware/physical address of a computer?
a. IP Address b. MAC Address c. Static IP d. Subnet Mask
2. Speed of data is expressed in:
a. Hz/s b. words/s c. bits/s d. bits/min
3. Which of the following is not an active network component?
a. Cable b. Router c. Hub d. Switch
4. Ping is used to
a. Assign IP b. Login c. Test a network d. search files
5. There are ____ classes of IP addresses.
a. 4 b. 3 c. 5 d. None of these

Write short answer of the following.

1. Define Computer Networks.
2. Differentiate between wired and wireless networks.
3. What is the difference between accesspoint and stations?
4. Define Router.
5. Define Switch.

Answer the following question in detail.

1. Differentiate between LAN and PAN.
2. Differentiate between IP address and MAC address.
3. Differentiate between IPv6 and IPv4.
4. Describe network topologies.
5. Describe DHCP mode.

Practical Tasks

1. Assign static IP addresses to the computers in the computer lab.
2. Perform Ping test to check the Internet connectivity of your computer.